

Contextual bandit against the rule: complete results

Protocol. One pull is one real request on the phone: Llama-3.2-1B on the Adreno 840, 9737-token prompt, 1024 output tokens, $K=1024$, μKV frozen config. Cooling gate ($\text{DDR} \leq 35^\circ\text{C}$, $\text{battery} \leq 33^\circ\text{C}$) before every pull, CPU caps pinned, GPU cap restored after each pull. Energy is metered from the USB rail plus the coulomb counter. Epsilon-greedy, $\epsilon = 0.25$, warm start of one pull per (context, arm). 24 pulls on 2026-09-03, 11:08 to 13:37.

Reward. $r = w_q - w_t T/T_{\text{ref}} - w_e E/E_{\text{ref}}$ with $E_{\text{ref}} = 782.0\text{J}$ and $T_{\text{ref}} = 142.4\text{s}$, the cost table’s prediction for this request at 1200 MHz. The weights are the scheduler’s own tier weights, so the bandit optimizes exactly the objective the rule is built to serve.

Table R1. Contexts and their weights. Identical to the scheduler’s lever anchors.

Context	battery	w_q	w_t	w_e	time slack
healthy	mains or >50%	0.55	0.40	0.05	5%
mid	21 to 50%	0.35	0.20	0.45	22%
low	$\leq 20\%$	0.20	0.10	0.70	40%

Table R2. Every context and arm the bandit pulled. Energy and time are measured means. The bandit’s preferred arm is marked. All 24 pulls are represented.

Context	Arm (MHz)	pulls	Energy (J)	Time (s)	Mean reward	
healthy	1200	6	804.6	139.1	+0.1079	bandit’s choice
healthy	902	1	599.4	180.6	+0.0043	
healthy	726	1	559.0	218.4	−0.0991	
mid	1200	1	799.5	139.1	−0.3054	bandit’s choice
mid	902	6	579.6	180.5	−0.2370	
mid	726	1	579.5	218.5	−0.2903	
low	1200	1	790.2	139.2	−0.6050	bandit’s choice within 0.0015 of 902
low	902	4	591.4	180.7	−0.4562	
low	726	3	563.4	218.4	−0.4577	

Table R3. What learning bought. Utility is the tier’s own objective evaluated on the cost table, so rule and bandit are scored on the same scale. The rule may pick split-clock plans the bandit’s arm set does not contain.

Context	Rule’s plan	Utility	Bandit’s arm	Utility	Gain	Verdict
healthy	prefill 1200, decode 902	+0.0989	1200	+0.0999	+0.0010	tie
mid	prefill 1200, decode 726	−0.2845	902	−0.2440	+0.0405	bandit wins
low	902	−0.4551	902	−0.4551	0	identical

Table R4. The one disagreement, in physical units. The rule refuses the bandit’s arm because it misses the mid tier’s time budget of 173.1 s, not because it ranks it lower.

Mid tier, 9737 prompt + 1024 output	Energy (J)	Time (s)	Utility
Rule: prefill 1200, decode 726	750.0	144.5	−0.2845
Bandit: flat 902	589.5	181.4	−0.2440
Difference	−160.5 (−21%)	+36.9 (+26%)	+0.0405 (+14%)

Table R5. What learning cost. Regret is measured against an oracle that always plays the bandit’s own final best arm. Almost all of it is paid in the mandatory warm start.

	Pulls	Energy (kJ)	Regret
Warm start, one pull per context and arm	9	5.9	0.542
Free choices	15	9.8	0.043
Total	24	15.7	0.585
Wall-clock cost including the cooling gates: 149 minutes			
The rule pays none of this. Its table is seeded from measurements taken once, offline.			